

ABSTRACT

Regional transmission organizations must match electricity supply and demand continuously, and day-ahead demand forecasts are least reliable exactly when the consequences of error are greatest. This project evaluated whether Adaptive Conditional Neural Processes (AdaCNP), a probabilistic few-shot method reported to improve extreme-load forecasting on the PJM and ISO-NE systems, reproduces that advantage on ERCOT data. A leave-one-event-out protocol was built over 17 load-eligible extreme cold events spanning 2002–2026, with plus/minus seven-day buffers excluded from both training and context retrieval, a 09:00 CT day-ahead issuance cutoff enforced throughout, and an adopted treatment for hours whose load-shed status is documented or unknown. The comparison against a standard Conditional Neural Process baseline was pre-registered: the analysis plan, primary endpoint, statistical test and decision rule were committed to version control before any confirmatory run executed. Two findings emerged. First, and most consequential for grid operations, both models are severely overconfident during extreme events. The nominal 90 percent prediction interval contains the observed load for 96 percent of normal-period hours but only 63 percent of held-out extreme-event hours, and coverage degrades further as events become more severe. The failure is directional: the models systematically under-forecast demand while reporting high confidence. Second, the pre-registered test detected an AdaCNP advantage on the primary metric, a mean paired difference of 0.459 in Gaussian negative log likelihood with 95 percent confidence interval 0.108 to 0.811 and p equal to 0.014 over 17 events. Three qualifications are reported alongside it: the observed effect lies just below the study’s minimum detectable effect of 0.495, so power at the realised effect size was approximately 77 percent; the advantage appears only in the pre-registered configuration; and the design cannot resolve an effect of the magnitude the source paper reports, which is roughly an order of magnitude smaller. A measurement of the proposed mechanism found that AdaCNP places less weight on cold historical analogues than uniform weighting does, so the performance gain is not explained by the mechanism its authors describe. The calibration deficit, rather than the architecture comparison, is the result most relevant to reserve planning and early generation commitment.

DISCLAIMER

This project was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS), under the Science Undergraduate Laboratory Internships (SULI) Program through an appointment at the National Energy Technology Laboratory (NETL). Neither the United States Government nor any agency thereof, nor any of its employees, nor the support contractor, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.”

Motivation Why accurate energy forecasting matters to the DOE mission • [Frame the problem: what decision depends on an energy forecast, and at what scale?] • [Who relies on it — grid operators, resource planners, DOE program offices?] • [What is the cost of being wrong: reliability, dollars, emissions?] • [One anchoring statistic or short real-world example] • [Connect explicitly to NETL / DOE mission relevance] 3

Background: Conventional Forecasting and Its Limits Where existing approaches fall short Current state of practice • [Name the standard methods used today — statistical, physics-based, or ML] • [What assumptions do they rely on?] Known limitations • [Limitation 1 — e.g. accuracy under volatile or extreme conditions] • [Limitation 2 — e.g. data requirements, computational cost, interpretability] • [The specific gap your project addresses] 4

Research Objectives The specific question this project set out to answer Research question • [State the question in one sentence a non-specialist could follow] Objectives 1. [Objective 1 — what you built, measured, or tested] 2. [Objective 2 — how you evaluated it] 3. [Objective 3 — what you compared it against] Scope • [What was in scope for a 10-week internship, and what was not] 5

Data and Methods What went in, what was used, and how it was evaluated Data Novel method • [Source(s) and provider] • [Name the method] • [Time span and resolution] • [What makes it novel here] • [Number of records / features] • [Cleaning and preprocessing steps] [Key hyperparameters or settings] • [Software and compute used] Evaluation • [Baseline model(s) for comparison] • [Metrics: e.g. RMSE, MAE, MAPE] • [Train / validation / test split] • [Validation strategy] 6

Approach End-to-end workflow from raw data to validated forecast Workflow Workflow diagram 1. [Acquire and clean input data] 2. [Engineer features / define inputs] 3. [Train the model] [Insert your pipeline or architecture diagram here — a single clear figure is worth more than a list] 4. [Tune and validate] 5. [Benchmark against baseline] 6. [Interpret and report] 7

Results I: Forecast Accuracy vs. Baseline PLACEHOLDER DATA — replace with your measured results Forecast Error by Horizon (lower is better) Baseline method Novel method 16 14.1 14 MAPE (%) 12 9.5 10 8 6 4 6.8 4.2 9.4 6.2 4.6 3.1 2 0 1-hour 6-hour 24-hour 7-day Forecast horizon 8

Results II: Predicted vs. Observed Demand PLACEHOLDER DATA — replace with your measured results Predicted vs. Observed Load, Representative Day Observed Novel method forecast Baseline forecast 04:00 08:00 16:00 900 800 Load (MW) 700 600 500 400 300 200 100 0 00:00 12:00 Hour of day 20:00 24:00 9

Discussion What the results mean — and what they do not Interpretation • [Why does the novel method outperform / underperform?] • [Which conditions drive the largest gains?] • [How do results compare to published work?] • [Practical significance for a grid operator or planner] Limitations • [Data coverage or quality constraints] • [Assumptions that may not generalize] • [Computational or runtime trade-offs] • [What a longer study would need to confirm] 10

Conclusions and Future Work Key takeaways from the summer and where the work goes next Key findings Impact [Finding 1 — the headline result, quantified] • [Finding 2] • [Who could use it and how] • [Finding 3] • [Broader mission relevance] • [What this enables for DOE / NETL] Future work • [Next experiment or dataset] • [Model refinement to pursue] • [Path to publication or deployment] 11

Acknowledgments Thank you — questions welcome With thanks to • [Mentor name] — NETL, for project guidance and mentorship • [Federal team supervisor / division director name] • [Colleagues, group members, or collaborators who contributed] Support • This work was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS), under the Science Undergraduate Laboratory Internships (SULI) program. Questions? [your.email@example.edu]

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I. INTRODUCTION

Electricity cannot be stored economically at grid scale, so a balancing authority must match generation to demand continuously. That obligation forces a decision under uncertainty every day: commit generating resources today for demand that will materialise tomorrow. Under-commit and reserves fall short; over-commit and consumers pay for capacity that idles. Regional transmission organizations and independent system operators are permitted to commit resources early, before the day-ahead market, in order to hedge extreme weather, but doing so runs against their mandate to dispatch at least cost. The value of that hedge depends entirely on whether the forecast’s stated uncertainty can be trusted.

Extreme weather is where this is hardest and where it matters most. The February 2021 winter storm drove ERCOT to shed firm load. Such events are also rare by construction, which creates a data problem: a model may have tens of thousands of ordinary days to learn from and only a handful of the days that actually determine reserve adequacy. Worse, the relationship between weather and demand is not stable across regimes, so ordinary days can actively mislead.

This project addressed that problem by testing a published method on new data. Hu *et al.*¹ propose Adaptive Conditional Neural Processes, which extend Conditional Neural Processes by weighting historical context examples according to their similarity to the current forecast target rather than averaging them uniformly. They report improved extreme-load forecasting on the PJM and ISO-NE systems. The goal of this project was to determine whether that reported advantage reproduces on ERCOT data, and to establish what such a test can and cannot demonstrate given the number of extreme events ERCOT’s history supplies.

Project boundaries were fixed at the outset. The comparison is AdaCNP against standard CNP only, on a shared encoder–decoder backbone, with the aggregation mechanism as the single intended difference; no other architectures were evaluated. Evaluation is restricted to held-out extreme cold events under a leave-one-event-out protocol. All work is retrospective on historical data; nothing was deployed operationally.

II. METHODOLOGY

A. Data and event definition

The target series is hourly ERCOT system load from 2002 to 2026, adopted by content hash rather than by filename so that provenance is exact. Extreme events come from a frozen inventory derived from a regional temperature index, with severity characterised by cold margin in degrees Celsius. Seventeen events are load-eligible, meaning they fall within the span of the load record; this count is derived from the artifacts at run time rather than hard-coded, so the protocol adapts if the inventory changes.

An episode targets one calendar day and predicts its 24-hour load vector. Inputs are calendar terms, the 24 most recent complete hourly load observations, the 24 most recent temperature observations, the 24-hour rolling mean temperature at the cutoff, and non-linear heating and cooling

terms: 57 features in total. Every feature is restricted to information complete at or before the day-ahead issuance instant, fixed at 09:00 America/Chicago on the day preceding the target day.

B. Leakage control and censored observations

Three protections were built and verified by automated test rather than by inspection. Outer partitions are leave-one-event-out with a seven-day buffer on each side of the held-out event, excluded from both training and context retrieval. Both model arms consume a single persisted context-index file, re-read from disk by each arm, so their context sets are byte-identical by construction. Normalization statistics are fitted on each fold’s training partition only.

Load measured during directed load shed is a lower bound on the demand that would otherwise have materialised, not an observation of it. Across the 1,271 event-hours in the record, 80 hours are documented as shed, zero hours are documented as not shed, and the remaining 1,191 carry no primary-source determination either way. The adopted treatment excludes documented shed hours from the primary metric, retains undetermined hours while flagging them, and reports an all-hours served-load likelihood as a secondary diagnostic on a different estimand. Resolving the undetermined hours in either direction would assert a finding the record does not support.

C. Statistical design and pre-registration

The primary metric is Gaussian negative log likelihood on held-out event-period load. Because normalization is fitted per fold, likelihood values are not comparable across folds; the only admissible quantity is the within-event paired difference between arms, which share a fold’s normalizer and context file. The unit of analysis is therefore one paired difference per event with the three frozen seeds averaged, giving a sample size of 17.

The analysis was pre-registered. The primary endpoint, the configuration to be tested, the statistical test, the significance level, the robustness checks and the decision rule, including what a null result would be permitted to claim, were committed to version control before any confirmatory run executed. The analysis code was committed at the same time and validated against earlier exploratory data, so the reported number was produced by code whose commit provably predates the data it analysed. The tested configuration was chosen for fidelity to Hu *et al.*—temperature features and randomly sampled context sets, matching their inputs and their sampling procedure—and explicitly not for the size of effect it had shown in exploratory work.

A power analysis preceded the confirmatory runs, using variance measured from exploratory data rather than assumed. It established the minimum detectable effect at 80 percent power, reported alongside every result so that a null can be distinguished from an absence of power.

The full sweep comprised 408 runs—17 folds by 2 arms by 2 feature sets by 2 context conditions by 3 seeds—executed on CPU with zero failures. Training length was not fixed by hand: each run trained to a ceiling and the reported parameters were those achieving the best inner-validation likelihood, with inner validation drawn from the fold’s own training partition. Selecting on the held-out event would have constituted leakage.

III. RESULTS AND DISCUSSION

A. Calibration fails during extreme events, in the dangerous direction

The most operationally consequential result is not the comparison between architectures. It is that both models' stated uncertainty is untrustworthy precisely in the regime the project set out to study.

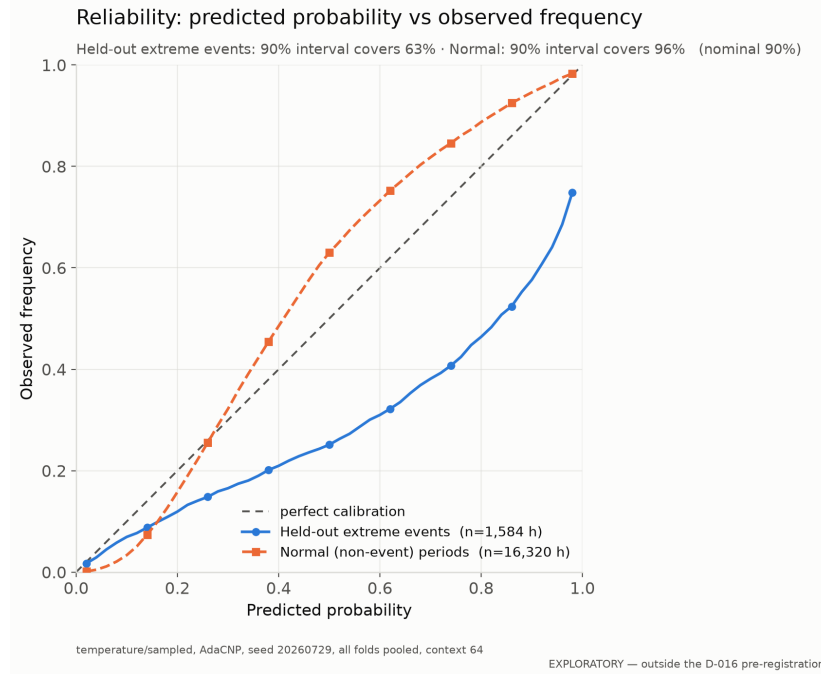


Figure 1: Reliability of the predictive distribution, by regime. For each nominal probability the model's corresponding quantile is computed and the observed frequency of outcomes at or below it is plotted; a perfectly calibrated forecast follows the diagonal. Both curves come from the same trained models, evaluated on held-out extreme-event days and on non-event days drawn from the same folds' inner-validation blocks. The 90 percent central interval contains the observed load for 96 percent of normal-period hours but only 63 percent of extreme-event hours. **Exploratory: outside the pre-registered analysis.**

Figure 1 shows the failure. In normal conditions the forecast is honest and slightly conservative, its intervals marginally too wide, which is the safe direction in which to err. During held-out extreme events the same models' 90 percent interval covers 63 percent of hours. The extreme-event curve also lies below the diagonal across its entire range, meaning fewer outcomes fall below each predicted quantile than the model expects: the models systematically under-forecast demand during extreme cold while reporting high confidence. This is the error mode that would lead an operator to skip an expensive early commitment.

Figure 2 makes the magnitude concrete for one day of the February 2021 event. Figure 3 shows that the deficit is not uniform: coverage declines as events become more severe, at approximately -0.037 per degree Celsius of cold margin for CNP ($r = -0.55$) and -0.031 for AdaCNP ($r = -0.51$). Only the mildest events approach nominal coverage. For practice, the implication is that calibration is worst exactly where reserve margins are thinnest.

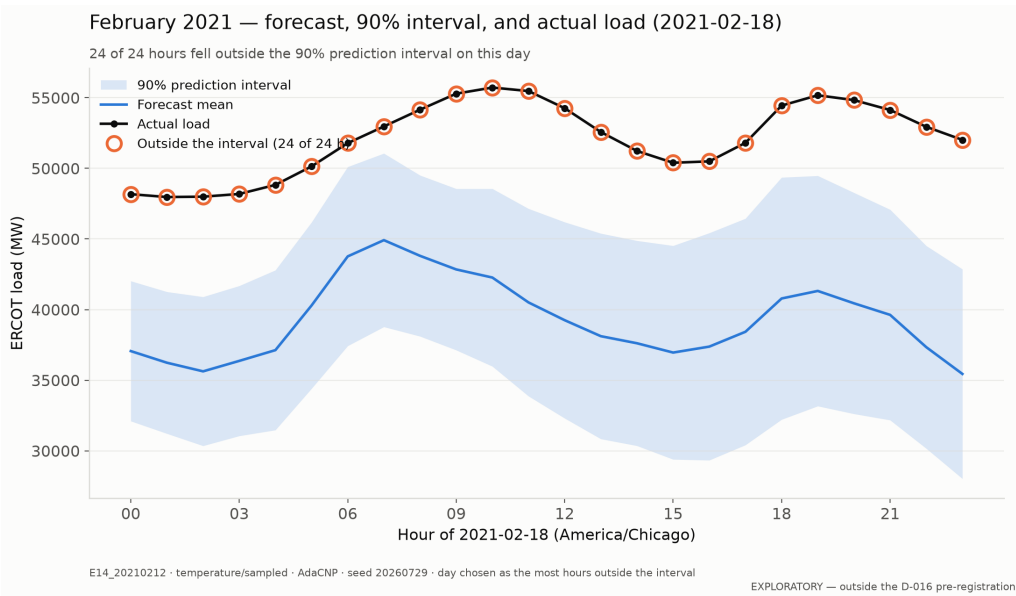


Figure 2: A single day of the February 2021 event. The day shown was selected by a stated rule—the day of that event with the most hours falling outside the 90 percent interval—and is therefore the worst case, not a typical day. All 24 hours fell outside the interval, with observed load running roughly 10 GW above the forecast mean. The day carries no documented load-shed hours, so the gap reflects forecast error rather than curtailed demand. **Exploratory.**

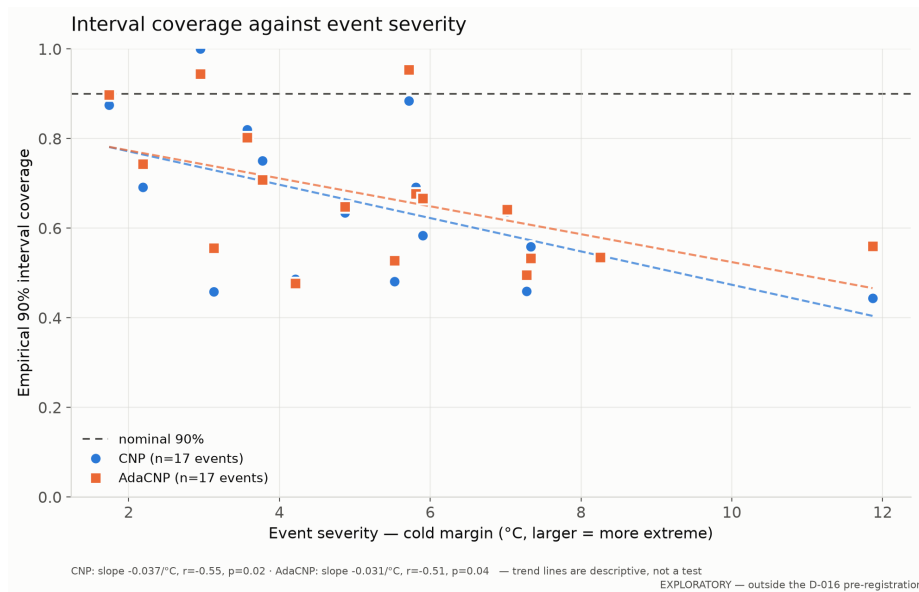


Figure 3: Empirical 90 percent interval coverage against event severity across all 17 leave-one-event-out folds, both arms. Trend lines are descriptive and form no part of the pre-registered analysis. **Exploratory.**

B. The pre-registered comparison

The pre-registered test detected an advantage for AdaCNP on held-out ERCOT extreme events; Table 1 reports it.

Table 1: Pre-registered primary result. Positive differences favour AdaCNP.

Quantity	Value
Events (n)	17
Seeds per event	3
Mean paired difference	+0.459
95% confidence interval	[+0.108, +0.811]
Two-sided paired t -test	$p = 0.014$
Wilcoxon signed-rank (robustness)	$p = 0.020$
Minimum detectable effect	0.495

Predeclared robustness checks are consistent. The Wilcoxon signed-rank test agrees with the t -test, so the pre-registered inconclusive branch was not triggered. Excluding the one fold reduced to a single held-out day gives a mean of +0.486 ($p = 0.014$). A leave-one-fold-out jackknife spans [+0.343, +0.501] with the sign stable across all 17 refits, so no single event drives the finding, and 11 of 17 events favour AdaCNP. Secondary metrics point the same way: continuous ranked probability score 0.469 against 0.486, root mean squared error 0.756 against 0.773, and calibration error 0.247 against 0.272. The frozen protocol reserves adjudication to the primary metric, so these are reported for interpretation only.

Three qualifications belong with that result rather than in a footnote.

First, the observed effect of +0.459 lies just below the minimum detectable effect of 0.495. The study therefore had roughly 77 percent power at the effect size it actually found, short of the 80 percent target. The result satisfies the pre-registered decision rule, but it sits near the edge of what this design can resolve and warrants independent replication rather than treatment as settled.

Second, the advantage appears only in the pre-registered configuration. Of the four configurations run, the pre-registered one gave +0.459; the same feature set with nearest-neighbour context retrieval gave -0.240 , favouring the baseline. Had the configuration been chosen after inspecting results, a range of conclusions could have been supported. The pre-registration is what makes the reported claim admissible, and that dependence is stated deliberately.

Third, this is not a measurement of the effect size Hu *et al.* report. Their AdaCNP-over-CNP likelihood margin corresponds to roughly 0.02 to 0.12 in the units used here, an order of magnitude below this design’s minimum detectable effect. The power analysis puts this study’s power to detect a margin that size at 5 to 9 percent and estimates that several hundred events would be required; ERCOT’s history supplies 17. A null would consequently have been uninformative about the source finding, which is precisely why the minimum detectable effect is reported beside the estimate.

C. The mechanism does not behave as described

AdaCNP’s proposed advantage comes from concentrating attention on relevant historical analogues, and during extreme cold that means similarly cold days. This is directly measurable

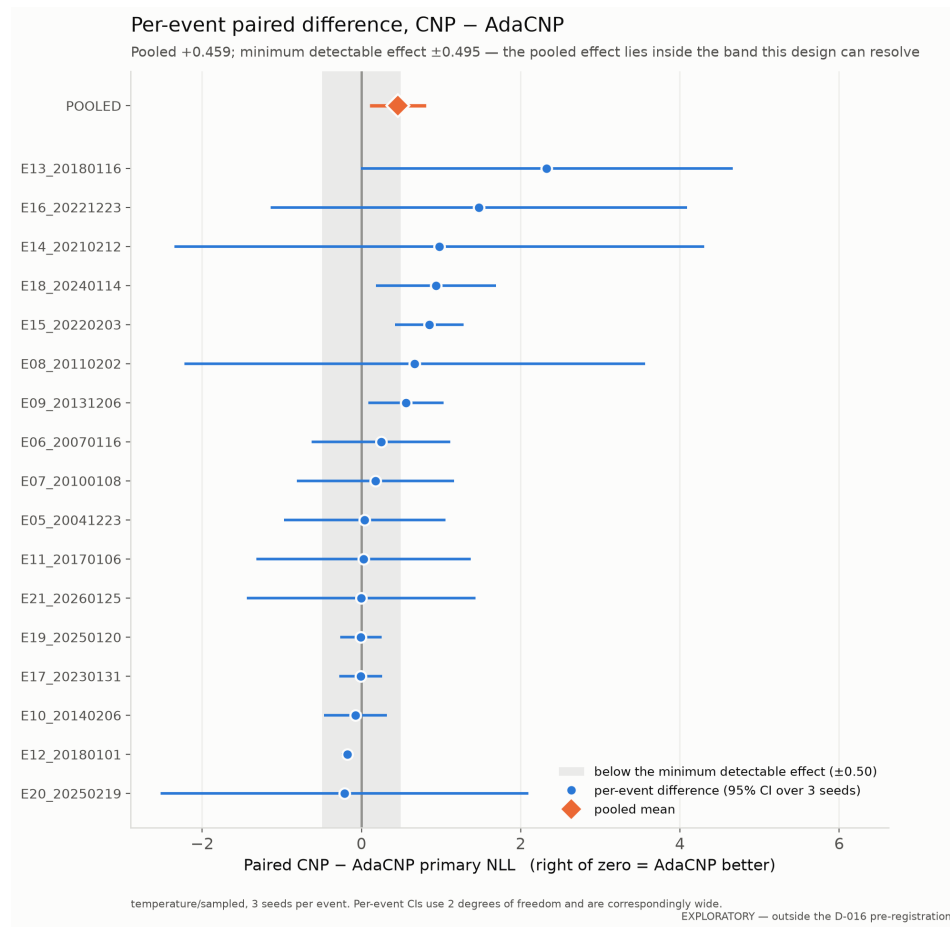


Figure 4: Per-event paired differences with 95 percent confidence intervals, the pooled mean, and the shaded band below the minimum detectable effect. Per-event intervals use two degrees of freedom and are correspondingly wide; they indicate shape, not per-event significance. **Exploratory presentation of the pre-registered quantity; the confirmatory result is Table 1.**

once temperature features are present. Under uniform weighting, CNP places 0.105 of its context weight on cold days, which is simply the cold fraction of its context set. AdaCNP places 0.079: it de-emphasises cold analogues, and its weighting is only mildly non-uniform, using 57.5 effective context points out of 64.

AdaCNP therefore scores better on held-out extreme events, but not by doing what its authors describe. Whatever produces the improvement observed here, it is not concentration on cold analogue days. This is an open question and, for future work, a more interesting one than the score difference itself.

D. Limitations

The event count is fixed at 17 by ERCOT’s history and by the extreme-weather framing; the power analysis identifies it as the binding constraint. Of the 1,584 held-out hours scored per seed, 1,139 carry undetermined load-shed status, so the primary metric carries an unknown amount of censoring contamination that the record cannot resolve. No day-ahead forecast weather product with historical issuance timestamps was available, so all temperature features are past-observed; Hu *et al.* additionally use next-day forecast temperature, and this remains the largest divergence from their protocol. Three seeds were used, initialisation noise dominates the measurement, and additional seeds would be the cheapest available improvement to precision. Finally, only the primary endpoint is confirmatory; the calibration results and the mechanism measurement are exploratory and labelled as such throughout.

IV. CONCLUSIONS

The pre-registered comparison detected an advantage for AdaCNP over a standard CNP baseline on held-out ERCOT extreme cold events: +0.459 in Gaussian negative log likelihood, 95 percent confidence interval [+0.108, +0.811], $p = 0.014$, $n = 17$, with robustness checks consistent and no single event driving the result. The advantage is real by criteria fixed in advance, but it sits near the resolution limit of the design, it depends on the pre-registered configuration, and it is not a measurement of the effect size reported in the source work.

The more consequential finding for grid operations is that both models are badly and directionally overconfident during extreme events. A nominal 90 percent interval that contains the outcome 96 percent of the time in normal conditions contains it 63 percent of the time during extreme cold, with the deficit widening as events worsen. For a balancing authority weighing early generation commitment, a forecast that is uncertain and says so is manageable; a forecast that is wrong and confident is not. On this evidence, improving calibration in the extreme regime would repay more than further architecture comparison.

A. Summary of appointment activities

Over the appointment I built a complete and reproducible experimental pipeline for probabilistic extreme-load forecasting on ERCOT data: verified data import with content-hash provenance, leakage-controlled leave-one-event-out partitioning with issuance-time enforcement, an adopted treatment for censored load observations, a CPU-based training and evaluation harness for both

model arms, a pre-registered statistical analysis, and a power analysis quantifying the design’s resolution. Activities included a literature review that identified the source method, reconstruction and hash verification of the project’s inherited data corpus, a documented audit comparing this implementation against the source publication, and execution of a 408-run confirmatory sweep. Decisions of record are captured in a versioned decision log, and every reported figure is regenerated from committed artifacts by script rather than transcribed.

B. Related activities and publications

[TO BE COMPLETED BY AUTHOR. NETL requires a summary of related activities (presentations, training, conferences, learning activities) and any publications. This is personal record and is not contained in the project repository. List the end-of-summer presentation, any training completed, any conferences or seminars attended, and any publications — or state “None.”]

ACKNOWLEDGMENTS

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REFERENCES

- [1] Chenxi Hu, Yue Ma, Yifan Wu, and Yunhe Hou. Resilient load forecasting under climate change: Adaptive conditional neural processes for few-shot extreme load forecasting, 2026. URL <https://arxiv.org/abs/2602.04609>.